

## Lumen Chart: EHR Clinical Documentation Redesign

**Project Overview:** Lumen Chart is a conceptual clinician-facing EHR redesign created for the Foster Healthcare UX class. This case study is grounded in real-world research, including heuristic evaluations, personas derived from clinical workflows, and competitive analysis. All research citations are included on the project page.

**Challenge:** Clinicians spend 6–7 hours per day in the EHR — more time charting than seeing patients. Existing systems suffer from alert fatigue, fragmented information, poor AI transparency, and high documentation burden, leading to burnout and safety risks.

### Goals:

- Reduce documentation time while maintaining safety and accuracy
- Improve AI transparency and clinician trust through clear provenance
- Support interrupted workflows with progressive disclosure and contextual tools
- Surface SDOH factors early without adding cognitive load

**Research & Insights:** Conducted heuristic evaluation (Nielsen + PAIR + HAX + NIST + WCAG), research-grounded personas, and competitive analysis of Epic, Cerner, and Athenahealth. Key findings: alert fatigue, fragmented documentation, and the need for deliberate AI review gates.

### Personas

- **Dr. Laura Mitchell** – Attending Physician (high skepticism of AI, alert fatigue)
- **Kevin Hartley, RN** – Charge Nurse (bedside constraints, real-time handoff needs)
- **Diane Santos** – Family Caregiver (plain-language needs, coordination across family)

### Key Design Decisions

- **Contextual Progressive Disclosure:** Only the most relevant information is shown initially; additional details expand on demand
- **Amber AI Diff Highlighting:** Clear visual distinction between AI-generated and verified chart content
- **Per-Section Accept Gates + Two-Gate Signing:** Forces deliberate review before permanent signing
- **Alert Hierarchy:** Critical alerts are visually distinct and actionable from the dashboard
- **SDOH Surfacing:** Visible at queue and snapshot level with linked resources

**Solution:** A high-fidelity interactive HTML prototype covering the full clinician workflow: smart daily dashboard, AI patient snapshot with SDOH panel, SOAP note editor with amber diffs and transcript toggle, and pre-sign confirmation with mandatory acknowledgments.

**Interactive Prototype:** [Live Interactive Prototype →](#)

A short interactive tutorial (based on user feedback) is included in the **Quick Actions panel on the right** to help visitors learn the key features and interactivity.

**Outcomes & Learnings:** Heuristic evaluation identified no catastrophic issues but revealed several usability concerns (e.g., missing Unaccept controls, weak error recovery paths, and pre-sign gate vulnerabilities). These issues are documented on the project page along with targeted solutions implemented in the prototype. A separate moderated think-aloud session con

The redesign demonstrates how progressive disclosure, provenance, and safety gates can meaningfully improve AI-assisted documentation in real clinical environments, though continued iteration is needed before production use.